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RISK-AWARE SLA ESTABLISHMENT IN AGREEMENT NETWORKS – AN EVALUATION

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Abstract

The rise of Software as a Service (SaaS) platforms is driven by the current trend towards customer-oriented, agile and scalable service provisioning. Customers' demand for highly specialized services can mostly not be met by a single provider that offers a holistic solution but by a consortium of providers that cooperate. In order to enable the combination of services and hence, the collaboration between different service providers, technical challenges like the compatibility of interfaces have to be dealt with. In addition, economic issues concerning profit, reliability and risk of service failure have to be taken into consideration.

A model of Agreement Networks (ANs) is presented that illustrate the legal bindings that are established or under negotiation between service providers and a methodology is introduced that supports providers in ANs to decide on the establishment of a combination of Service Level Agreements (SLAs) that incurs the lowest risk of SLA violation. Finally, a benchmark for risk-based decisions is derived that serves to evaluate the efficiency of the choice. The impact of the available amount of monitoring data on the decision is investigated and a minimal amount of 100 observations for each SLA combination is suggested.

Keywords: Service Composition, Service Level Agreement, Agreement Network, Risk.

1 Introduction

The service market changed remarkably over the last years. From monolithic, holistic solutions that were offered “on premise“ by single providers, customer requests are nowadays met by composing a variety of standardized services. This trend manifests itself in a growing demand for web service compositions.¹

In order to facilitate the dynamic composition of services in highly business-relevant scenarios, agreements on the quality and other terms of provision of the respective services are established. While existing literature on service composition mainly focuses on technical feasibility or on the efficient choice of service providers for singular service requests, this work illustrates the interdependent network of Service Level Agreements (SLAs) that arises between service providers. Agreement Networks (ANs) depict providers and their legal bindings that are established or under negotiation. Those bindings include service offers enriched by quality assurances as well as prices for service execution and penalties that have to be paid in case of (partial) service failure. In order to enforce the application of penalties, service monitoring has to be in place.

A service provider who offers services in an AN is concerned with the adherence to active SLAs and wants to avoid SLA violations. SLA violations lead to the payment of penalties. However, public knowledge about the violation of an SLA can lead to a loss of reputation and consequently could decrease future demand for the provider’s services. Hence, a provider in an AN has to account for their risk of violating an SLA when deciding about the establishment of SLAs. This paper presents an approach that allows for risk-based decision about the establishment of combinations of SLAs about services that a provider offers.

In order to substantiate the theoretical results, a benchmark for risk-based decisions derived from known probability distributions of SLA violations is provided. Furthermore, the impact of the amount of available monitoring information, that is, the available number of observations is investigated and it is shown that a relatively low amount of monitoring data is required for a high degree of efficient decisions.

This paper is structured in seven sections. Section 2 gives an overview over related literature and discusses complementary work. In Section 3, the need for risk-based decisions in agreement networks is motivated by means of an outsourcing scenario of highly business-relevant services. Section 4 introduces a methodology that supports decisions about the establishment of risk-minimal SLA combinations. In Section 5, an approach is introduced that allows the benchmarking of the methodology introduced in Section 4. The results of the evaluation are presented in Section 6. Section 7 concludes the paper and discusses further challenges and future work.

2 Related Work

The trend of business cooperation away from static value chains towards networks of firms has been observed for years. The rise of so-called Business Webs (see Steiner (2004); Tapscott, Lowy and Ticoll (2000)), where firms provide complements to a common product architecture, builds the basis for the composition of services. The technical composition of Web services that facilitates the provision of automated complex services such as insurance brokering, travel planning, insurance liability services or package tracking has been discussed intensively in the literature (Papazoglou and Dubray 2004).

¹ <http://www.programmableweb.com/> [last accessed: December 2, 2011]

Apart from the technical perspective of service composition, economic aspects like the coordination of value creation are of major concern. In Blau 2009, a mechanism-design approach is applied in order to find the efficient combination of service offers that on the one hand satisfies a customer's preferences and on the other hand meets providers' constraints. This mechanism distributes the customer's payment to those providers that are chosen in the efficient allocation. It considers the possibility of service failure and penalizes providers appropriately. Incentives for providers are set to submit truthful offers about a service's properties. The risk of service failure is not taken into account *ex ante* when choosing the efficient combination of service offers but is only penalized *ex post*, that is, after service execution. A different approach is chosen in Conte, et al. 2009 that relies on work from cooperative game theory. The payoff function in Conte, et al. 2009 distributes the customer's payment contribution-based to all providers that are offering their services and hence, contribute to the variety of alternatives. The risk of service failure is not taken into account.

A model that describes governance, risk and compliance for service outsourcing is introduced in Asnar, Lim, et al. (2010). Challenges in the context of Service Oriented Architectures (SOAs) and hence, service composition are highlighted and the paradigm of compliance by control is applied for enforcing compliance based on monitoring results. The model allows for the qualitative identification of risk, that is, what the risk is, and the definition of counter-measures in a very technical way but it does not support the quantitative measurement of risk, how severe the impact of the risk is, or risk-based decision support. In Asnar, Giorgini and Mulyanto (2007), a risk management model is introduced. Analogously to Asnar, Lim, et al. (2010), the identification of risk is possible, but quantitative risk measurement is not possible with this approach. Similarly, Kompella, Yates and Greenberg (2010) introduce an approach for risk modeling for localizing fault in IP networks. Risk is applied for the analysis of monitoring data to identify common failure patterns. This way, error propagation along links in IP networks can be detected and common risks can be identified. Analogously to Asnar, Giorgini and Mulyanto (2007) and Asnar, Lim, et al. (2010), a qualitative description of risk is given and no quantification or metric is introduced that would allow the evaluation of the severity of a failure.

In summary, the aforementioned literature either proposes an efficient allocation but does not take service failure into account *ex ante* or does not give the possibility to quantify risk for decision support. Michalk, et al. (2010) introduce a decision support model for choosing the risk-minimal SLA combination to establish. The model allows for risk stemming from past SLA violations with respect to concurrently established SLAs. The approach is based on H. M. Markowitz (1952) and H. M. Markowitz (1959), where the risk of security portfolios is determined by means of past variations of the portfolios' return. In order to be able to express SLA violations quantitatively, Michalk, et al. (2010) assume the existence of a resource monitoring system that creates reports about the relative adherence to a portfolio of SLAs. In contrast, the work at hand relies on the assumption that a service monitoring system is in place, so that each SLA is monitored individually. For expressing risk, the concurrently established SLAs are taken into account. The approach presented in Sections 3 and 4 adapts the risk-based decision model from Michalk, et al. (2010) for a scenario, when service monitoring is in place. Finally in Sections 5 and 6, the approach is evaluated.

3 Scenario

Service composition is facilitated through the establishment of agreements - commonly called SLAs. An SLA is specified according to Andrieux, Czajkowski and Keahey (2007) by the provider p , the consumer $\theta \in \Theta$, the respective service to be executed, $s \in S$, as well as multiple Service Level Objectives (SLOs) $l_\alpha \in L_\alpha$. SLOs are constituted of key performance indicators (KPIs) that are defined as target values of measurable properties of a service execution, e.g. response time in *ms*, throughput in GB/s, etc. Furthermore, an SLA α comprises a price $f_\alpha(s)$ that is charged for service execution as well as a penalty, μ_α , that the provider has to pay in case of an SLA violation. An SLA α is therefore defined as $\alpha = (p, s, \theta, L_\alpha, f_\alpha(s), \mu_\alpha)$.

Service providers are characterized by their supply of services $s \in S$ as well as their resource constraints L_p that denote the maximum amount of available technical resources such as disk space, throughput rate or availability. Note that resource constraints do not map one-to-one to KPIs that are defined in SLAs. Only those KPIs that relate directly to monitored resources can be applied when determining the violation of a resource constraint. Each service provider can have multiple SLAs even with the same contractual partner. The number and kind of contracts is only limited by the resource constraints L_p associated with the service provider p .

The structure that results from agreements between service providers is depicted in Figure 1. An AN opens the perspective on the legal structure that facilitates the cooperation between service providers in contrast to the technical compatibility and provision of complex services. Figure 1 shows an AN that contains service providers p_1, p_2, p_3 and the customer θ .

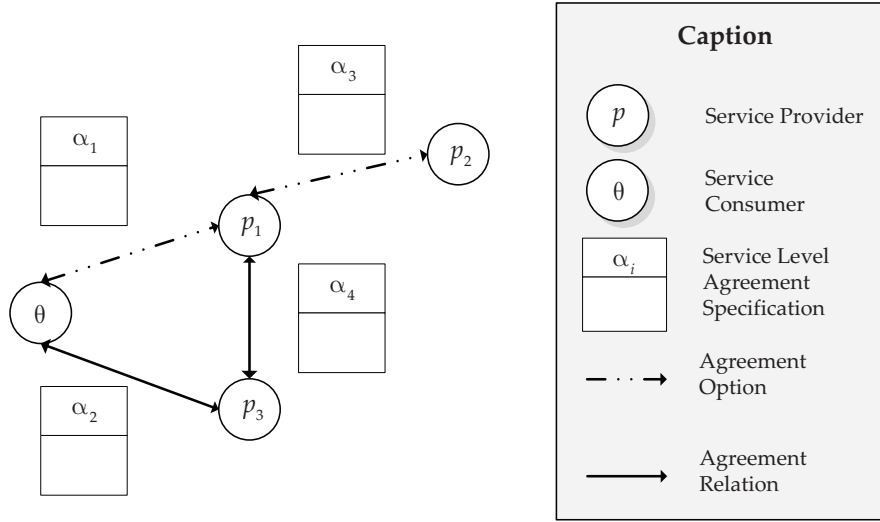


Figure 1. Agreement Network Extract

The decision problem that a service provider p is confronted with in an AN may be formulated as follows: What is the optimal configuration and combination of SLAs to be established?

In order to separate the effect of negotiation and strategic pricing decisions from SLA portfolio optimization, the configuration, that is, KPIs, prices and penalties, is assumed to be determined exogenously. The separation of effects in this work allows the analysis of risk-based decision making while avoiding the influence of pricing strategies on decision-making.

The availability of a data basis that is big enough to draw conclusions from is ensured by the restriction of the set of possible SLAs and their configurations to a predefined set A from which a service provider p may select.

A service provider p that has to choose an SLA portfolio γ knows that dependencies exist among KPIs. Exemplarily, assured throughput may influence the response time. As these dependencies are very complex to detect and to monitor in practice, the model at hand will not constitute dependencies explicitly. In theory, the straightforward approach to calculating the failure risk of a certain combination of SLAs would require the knowledge of the variances of failure of the single SLAs, as well as of the covariances between all possible SLAs. Applying this approach, however, would imply the assumption that the risk of SLA violation for one SLA is the same regardless of other concurrently active SLAs and hence, executed services. In the approach at hand, dependencies are incorporated implicitly by means of treating each possible combination of SLAs as a single entity, which *directly* reflects the correlation among attributes in the respective combination. The evaluation of the risk of failure is crucial for p as outages or failures will result in penalties μ that p has to pay in order to compensate the customers. Accordingly, a service provider's risk measure in ANs has to account not only for the probability and extent of failure but also for the size of the penalties that will result from a failure.

We assume that service providers' attitude towards risk can be represented by a von Neumann-

Morgenstern utility function that implies risk-aversion. According to Sandmo (1971), risk-aversion may be assumed for firms. Apart from avoiding risk, service providers are interested in as high an expected profit as possible through the provision of services.

A decision model is proposed that determines the optimal combination of SLAs γ for service provider p . Here, optimal means minimizing the risk imposed on p while considering a constraint on the expected profit, which specifies the minimum payoff the service provider wants to achieve. Furthermore, the resource constraints that describe the maximum resources available to the service provider are included in the decision problem. The following section introduces the optimization problem.

4 Methodology

As stated in Section 2, a service provider monitors their own adherence to single SLAs in portfolios γ and stores the collected data as periodic reports. The reports contain information about the agreements that were established in each period and their respective degree of violation. We assume the existence of a metric that aggregates the degree of violation of single KPIs into one value $\lambda_{\gamma,t}^{\alpha_i}$ that states relative violation of an SLA. In contrast to the approach in Michalk, et al. (2010) that relies on resource monitoring, in this paper service monitoring is applied. This implies that each SLA is monitored individually and a degree of violation can be determined respectively.

A sample of a provider's monitoring data including the periods t , the respective agreements for this period α_i and relative failure of an SLA $\lambda_{\gamma,t}^{\alpha_i}$ is shown in Table 1.

Time	SLA α_1	SLA α_2	SLA α_3
1	0	0	
2	0	0	
3		0.5	0.5
4	1	1	
5		0.7	0.4
6	0.78	0.9	0.8
7	0.82	0.9	
8	0.67		0.5
9	?	?	?

Table 1. Extract of Service Provider's Monitoring Reports

The entries in the first row of Table 1 show that in $t = 1$ the service provider established two different SLAs, α_1 and α_2 ($\gamma = \alpha_1, \alpha_2$), and he failed on none of them ($\lambda_{\{\alpha_1, \alpha_2\},1}^{\alpha_1} = 0, \lambda_{\{\alpha_1, \alpha_2\},1}^{\alpha_2} = 0$).

Providers' risk in the focus of this paper stems from the uncertainty of SLA violation and the resulting monetary outcomes. Applying the semi-variance (H. M. Markowitz 1991) of penalties as a proxy for risk implies that those violations of SLAs that lead to higher penalties than expected are perceived as uncertain and hence, are taken into account for the calculation of risk.

The semi-variance of penalties as a proxy for risk has been employed in portfolio optimization ever since the beginnings of finance theory and is especially useful to account for asymmetric probabilistic distributions and downside risk.²

In order to apply the semi-variance as a proxy for risk, the following assumptions need to hold:

- (1) Probability distributions of SLA violations don't change over time.

² See Levy (1992) and Eeckhoudt, Schlesinger and Gollier (2005), for an overview of the different risk measures and a definition of downside risk aversion as well as Elton, et al. (2003) and H. M. Markowitz (1959) for its relation to the shape of the distribution function.

- (2) The aggregation of relative violations of single KPIs to one SLA violation value is possible.
- (3) Each provider can compare violation values of different SLAs (a violation of 0.5 for SLA 1 implies the same as a violation of 0.5 for SLA 2).

A provider p can apply the semi-variance as a proxy for risk for an SLA combination γ if at some point in the future the same combination of SLAs γ is chosen. The calculation of the semi-variance of penalties for a certain SLA combination γ is conducted in four steps:

1. the actual sum of penalties $\mu_{\gamma,t}$ for each period is calculated,
2. the mean percentage of failure $\bar{\lambda}_{\gamma}^{\alpha_i}$ for a particular SLA α_i belonging to a specified SLA configuration γ is computed,
3. the average penalty of γ that p was supposed to pay is calculated based on $\bar{\lambda}_{\gamma}^{\alpha_i}$ and finally
4. the semi-variance is calculated as the squared deviations of actual penalties from the mean penalties.

The actual sum of penalties $\mu_{\gamma,t}$ that a service provider would have had to pay in period t depends on the relative failure of SLA α_i in the combination of SLAs γ in this period and is given by:

$$\mu_{\gamma,t} = \sum_{\alpha_i \in \gamma} \lambda_{\gamma,t}^{\alpha_i} \cdot \mu_{\alpha_i} \quad (1)$$

Define $X(\lambda_{\gamma,t}^{\alpha_i}) = 1$ if $\lambda_{\gamma,t}^{\alpha_i} \in [0,1]$ and $X(\lambda_{\gamma,t}^{\alpha_i}) = 0$, otherwise. This variable serves to indicate whether the SLA α_i was established ($X(\lambda_{\gamma,t}^{\alpha_i}) = 1$) or not ($X(\lambda_{\gamma,t}^{\alpha_i}) = 0$) in a particular period t .

The mean percentage of failure for a particular SLA α_i belonging to a specified SLA configuration γ may be calculated as

$$\bar{\lambda}_{\gamma}^{\alpha_i} = \left\{ \frac{\sum_t \lambda_{\gamma,t}^{\alpha_i}}{\#\lambda_{\gamma,t}^{\alpha_i}} \mid X(\lambda_{\gamma,t}^A) = 1 \wedge X(\lambda_{\gamma,t}^{\gamma^c}) = 0 \right\} \quad (2)$$

where $X(\lambda_{\gamma,t}^A) = \prod_{\alpha_i \in \gamma} X(\lambda_{\gamma,t}^{\alpha_i})$ and $X(\lambda_{\gamma,t}^{\gamma^c}) = \prod_{\alpha_i \in \gamma^c} X(\lambda_{\gamma,t}^{\alpha_i})$, with γ^c referring to all SLA portfolios other than γ . The indicator variables $X(\lambda_{\gamma,t}^A)$ and $X(\lambda_{\gamma,t}^{\gamma^c})$ serve to select from the observations in the service provider's records only the subset of observations, which pertains to a particular portfolio of SLAs γ . Thus, $\#\lambda_{\gamma,t}^{\alpha_i}$ is the number of periods in that the particular SLA combination γ appears in the records of the service provider.

For the average penalty, which the service provider was supposed to pay in the past for the combination of SLAs γ , one obtains

$$E(\mu_{\gamma,t}) = \sum_{\alpha_i \in \gamma} \bar{\lambda}_{\gamma}^{\alpha_i} \cdot \mu_{\alpha_i} \quad (3)$$

Based on this, the semi-variance of due penalties for the portfolio of SLAs γ can be calculated. It is the risk measure based on which the service provider chooses the optimal combination of SLAs and it is defined as:

$$S_E(\mu_{\gamma,t}) = \left\{ \sum_t q_t \cdot \left((\mu_{\gamma,t} - E(\mu_{\gamma,t}))^+ \right)^2 \mid X(\lambda_{\gamma,t}^A) = 1 \wedge X(\lambda_{\gamma,t}^{\gamma^c}) = 0 \right\} \quad (4)$$

$$\text{where } q_t = \frac{1}{\#X(\lambda_{\gamma,t}^A)=1} \text{ and } (\mu_{\gamma,t} - E(\mu_{\gamma,t}))^+ = \begin{cases} 0, & \mu_{\gamma,t} - E(\mu_{\gamma,t}) \leq 0 \\ \mu_{\gamma,t} - E(\mu_{\gamma,t}), & \text{else} \end{cases}$$

In order to represent the decision model as stated in Section 3, apart from the objective function (Equation (4)), the constraints on available resources of the provider L_p and the minimum expected profit requirement Π_p need to be defined:

$$\sum_{\alpha_i \in A} L_{\alpha_i} \cdot X(\lambda_{\gamma,t}^{\alpha_i}) \leq L_p \quad (5)$$

$$E(\pi_{\gamma,t}) \geq \Pi_p, \quad (6)$$

where $L_{\alpha_i} = (l_{\alpha_i}^1, \dots, l_{\alpha_i}^M)$ describes the KPIs (where M is the number of attributes) agreed upon in SLA $\alpha_i \in A$, and L_p analogously denotes the maximum of resources available to the provider.

The minimum expected profit the provider wants to achieve is denoted by Π_p , and the expected profit from an SLA combination γ is denoted by:

$$E(\pi_{\gamma,t}) = \sum_{\alpha_i \in A} (f_{\alpha_i}(s_j) - c(s_j)) \cdot X(\lambda_{\gamma,t}^{\alpha_i}) - E(\mu_{\gamma,t}), \quad (7)$$

where $f_{\alpha_i}(s_j)$ denotes the price that the provider charges for the provision of Service s_j in SLA α_i and $c(s_j)$ denotes the provider-specific costs that arise with the execution of service s_j .

In order to illustrate the approach consider the example in Table 1. For completeness of the numerical example further assume that the SLAs appearing in the table are defined as follows:

$$\alpha_1 \in A: (p_1, s_1, \theta_1, T_{put} = 5, f_{\alpha_1}(s_1) = 4, \mu_{\alpha_1} = 2)$$

$$\alpha_2 \in A: (p_1, s_2, \theta_2, T_{put} = 10, f_{\alpha_2}(s_2) = 11, \mu_{\alpha_2} = 2)$$

$$\alpha_3 \in A: (p_1, s_1, \theta_3, T_{put} = 6, f_{\alpha_3}(s_1) = 4, \mu_{\alpha_3} = 3),$$

where T_{put} denotes throughput in Mbit/s for the provision of the service. Let the provider p_1 face a maximum throughput of 15, which constitutes the optimization constraint (Equation (5)), and a minimum expected profit $\Pi_p = 5$ in money units (MU), as well as internal costs for service execution of $c(s_1) = 2MU, c(s_2) = 3MU$.

For the selection of the optimal SLA combination, the constraints have to be checked for each SLA combination and in case they are met the semi-variance is computed. Exemplarily, consider the SLA combination $\gamma = \alpha_1, \alpha_2$. It can be easily verified that this combination of SLAs satisfies the throughput constraint, as $l_{\alpha_1} + l_{\alpha_2} = 5 + 10 \leq L_p = 15$. In the past records it appears four times in periods $t \in \{1, 2, 4, 7\}$ such that $q_t = \frac{1}{4}$ in Equation (4).

The sum of penalties that the provider had to pay in each of these periods according to Equation (1) and the sample data from Table 1, is

$$\mu_{\gamma,1} = \mu_{\gamma,2} = 0 \cdot 2 + 0 \cdot 2 = 0, \mu_{\gamma,4} = 1 \cdot 2 + 1 \cdot 2 = 4, \mu_{\gamma,7} = 0.82 \cdot 2 + 0.9 \cdot 2 = 3.44$$

For the mean percentage of failure of the SLAs one obtains

$$\bar{\lambda}_{\gamma}^{\alpha_1} = \frac{0 + 0 + 1 + 0.82}{4} = 0.455, \bar{\lambda}_{\gamma}^{\alpha_2} = \frac{0 + 0 + 1 + 0.9}{4} = 0.475.$$

Insert these results into Equation (3), the average penalty paid in the past for the SLA combination γ is

$$E(\mu_{\gamma,t}) = \bar{\lambda}_{\gamma}^{\alpha_1} \cdot \mu_{\alpha_1} + \bar{\lambda}_{\gamma}^{\alpha_2} \cdot \mu_{\alpha_2} = 1.86$$

Substituting into Equation (7), it shown that the expected profit constraint (Equation (6)) is met:

$$E(\pi_{\gamma,t}) = \sum_{\alpha_i \in \gamma} (f_{\alpha_i}(s_j) - c(s_j)) - E(\mu_{\gamma,t}) = 8.14$$

After substituting into Equation (4) the semi-variance can be calculated.

$$S_E(\mu_{\gamma,t}) = \frac{1}{4} ((\max\{0; 0 - 1.86\})^2 + (\max\{0; 0 - 1.86\})^2 + (\max\{0; 4 - 1.86\})^2 + (\max\{0; 3.44 - 1.86\})^2) = 1.769$$

In order to select the risk-minimal combination of SLAs that satisfies the constraints in Equations (5) and (6), the above procedure has to be repeated for each SLA combination for which monitoring data is available. In this example, $\gamma = \alpha_1, \alpha_2$ is the only portfolio that meets the constraints. For single SLAs, there is no monitoring data available. $\gamma = \alpha_2, \alpha_3$ and $\gamma = \alpha_1, \alpha_2, \alpha_3$ violate the throughput constraint, whereas $\gamma = \alpha_1, \alpha_3$ violates the expected profit constraint. Therefore, the semi-variance only has to be calculated for $\gamma = \alpha_1, \alpha_2$.

5 Evaluation Plan

In a real-world scenario, a decision-maker takes into consideration all of the monitoring data that is available and comes to a decision based on this data. If there is no monitoring data for a specific service/SLA, the risk cannot be estimated by means of the semi-variance. However, in a real-world scenario, a provider will most likely have monitoring data for services that he offers.

The amount of available data has a major impact on the quality of the proxy that is calculated. The more monitoring data is available the more information is available on the distribution of SLA violations in certain SLA portfolios and hence, the better the proxy calculated from the data is. In order to find a threshold for the number of observations that is required for a sufficiently good decision, a benchmark for the decision has to be defined.

A straightforward approach for benchmarking the method would involve real monitoring data. The decision that has been made based on a certain number n of observations from the real monitoring data would be compared to the real SLA violation in period $n + 1$. Repeating this approach for different lengths of observations would lead to the desired threshold. As was argued in Section 4, the calculation of the semi-variance relies on assumptions that need to be met. Assumption (2) expects that the monitoring results are available as one aggregated value per SLA that specifies the degree of violation of the respective SLA. In order to enable the aggregation to one value per SLA, the degree of violation or adherence has to be specified for each SLO in the SLA. Furthermore, the decision-maker's preferences with respect to the impact of each of the SLOs have to be known. For most monitoring systems, monitoring reports contain only information the violation of KPIs in the form of Boolean values (yes/no) and not in the form of relative violation (e.g., x% violated). Aggregating this information, in case the decision-maker's preferences are known, would result in a value that states the relative violation of the SLA. A more precise proxy of relative SLA violation would require monitoring systems to determine the relative violation per KPI. Assumption (1) requires that probability distributions of SLA violations don't change over time. For real services that are monitored, this assumption cannot be met due to the fact that underlying infrastructure is subject to improvements and changes over time. Hence, real monitoring data can hardly be applied for benchmarking.

In order to overcome the drawbacks of real data, monitoring data is created by means of a numerical simulation that was implemented in Java. Benchmarking by means of simulated violation data is advantageous, as the probability distribution function (PDF) of SLA violations is known for all SLAs in all possible SLA combinations. The simulation is conducted as follows.

First, a predefined number of SLAs including prices and penalties is created. Then, all possible combinations of SLAs is created by means of the power-set of SLAs, and the violation probability distribution function (PDF) for the degree of violation is determined for each SLA in each SLA portfolio. Violations can be observed in $[0;1]$ and hence, a PDF with domain $[0;1]$ is required. Furthermore, parameterization of the PDF may not exceed a reasonable level, as the simulation is repeated for different instantiations of the PDFs. A family of distribution functions that meets those requirements is the family of Beta distribution functions, $B(a, b)$. Keeping b fixed and altering a in a specified interval allows to model a different PDF for an SLA depending on the concurrently active SLAs. The higher a is, the higher is the probability for a high degree of violations. The parameter b was arbitrarily chosen to be 1. Choosing a different number would have no impact on the shape of the distribution as long as the manipulation of a would stay the same. a is determined as follows:

$a = 0 + \#SLAs * x$, where x is a random draw in $U(0,1.0)$. In order to neglect effects that stem from the variation of prices or penalties the input parameters for number of SLAs, prices, penalties and costs were fixed. Table 2 summarizes the input parameters to the evaluation.

In the next step, the benchmark for the decisions based on monitoring data is calculated from the PDFs that were determined in the step before. Calculating the semi-variance of due penalties from the PDFs represents the most precise specification of risk and hence, is chosen as benchmark. The semi-variance is calculated by adapting the calculation of the variance of a PDF so that it takes only those deviations into account that are worse than the expected value. Additionally, the PDF is weighted by the penalties from the respective SLA. Consequently, the semi-variance of due penalties for combinations of SLAs is calculated as follows. For all SLAs in the portfolio, the sum of squared deviations from the expected penalty is calculated. The integral from the overall expected degree of violation through 1 of this sum determines the semi-variance of due penalties, as only those deviations that are worse than expected are taken into account:

$$\int_{expFailure}^1 (\sum_{\alpha_i \in \gamma} \mu_{\alpha_i} \cdot \tilde{\lambda} - \hat{E}(\tilde{\lambda} \cdot \mu_{\alpha_i}))^2 \cdot \sum_{\alpha_i \in \gamma} g_{\alpha_i}(\tilde{\lambda}) d\tilde{\lambda}, \quad (8)$$

where $g_{\alpha_i}(\tilde{\lambda})$ denotes the probability density function of violations of SLA $\alpha_i \in \gamma$, $\hat{E}(\tilde{\lambda} \cdot \mu_{\alpha_i})$ denotes the expected penalty, that is, the expected violations of the SLA portfolio γ weighted by the respective penalties μ_{α_i} .

Parameter	Value
Number of SLAs	3
PDF a	$0 + \#SLAs * x$
PDF b	1
Prices	$f_{\alpha_1}(s_1)=1.5, f_{\alpha_2}(s_2)=2, f_{\alpha_3}(s_3)=5$
Penalties	$\mu_{\alpha_1} = 2, \mu_{\alpha_2} = 1.3, \mu_{\alpha_3} = 5$
Costs	$c(s_1) = 1, c(s_2) = 1.5, c(s_3) = 2$
x	$x \sim U(0,1.0)$
n	1000
y	5,10,20,30,50,100,200,300,400,500,600,700,800,900,1000

Table 2 Evaluation Parameters

In the next step, samples of length y are drawn from the PDFs for each SLA in each portfolio. This way, for each portfolio, monitoring data of length y is obtained. In order to remove the statistical error from which computer-created random numbers suffer (Chung 2004), this random draw is repeated $n = 1000$ times.

For the risk-based decision, for each of the obtained monitoring samples, the semi-variance is calculated according to Equation (8). Based on this semi-variance from samples, the risk-minimal portfolio is chosen for each sample. Figure 2 illustrates the simulation setup.

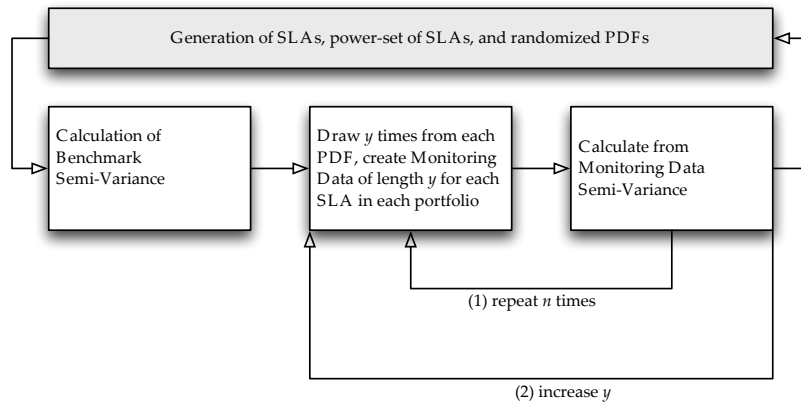


Figure 2 Simulation Setup

For each of the samples, the correctness of the sample-based decision is determined by comparing the chosen portfolio from sample data to the benchmark portfolio. The resulting value for correctness is stored in a table, so that there is one table for each of the distributions that shows for each length of history as illustrated in Table 2 for each of the n repetitions if the decision was correct or not.

The procedure from Figure 2 is repeated 150 times in order to achieve independence from a specific PDF. After running the Simulation for 150 times, mean fractions of correct choices with respect to different lengths of history as shown in Table 3 and Figure 3 are retrieved.

Length of History	Mean	N	Standard Deviation
5	.57816	150	.235697
10	.66289	150	.235654
20	.72848	150	.235191
30	.75949	150	.235105
50	.78821	150	.235929
100	.81622	150	.241372
200	.83239	150	.256013

300	.83936	150	.265794
400	.84352	150	.273200
500	.84645	150	.278414
600	.84889	150	.282153
700	.85066	150	.284915
800	.85214	150	.287201
900	.85371	150	.288685
1000	.85373	150	.291605
Total	.79695	2250	.273777

Table 3 Mean Fraction of Correct Choices for 150 Simulation Runs

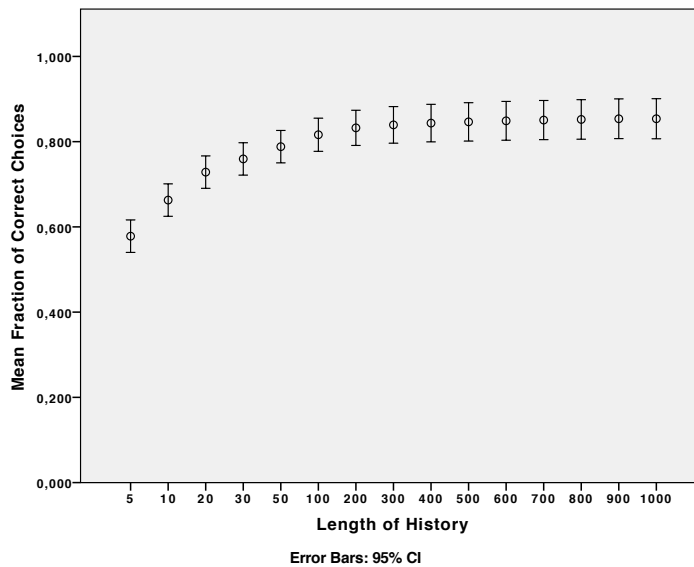


Figure 3 Mean Fraction of Correct Choices

The following Section illustrates the results of the statistical test for the required amount of monitoring data for a sufficiently good decision.

6 Results

The amount of required data is determined based on a statistical analysis as the threshold value from which on the fraction of correct choices cannot be increased significantly any more. Therefore, an analysis of variance is carried out in order to prove that the length of history has a statistically significant impact on the quality of the fraction of correct decisions. The hypothesis that the mean of the fraction of correct choices is equal for different lengths of history is analyzed. Afterwards, a two-sided matched-pairs t-test is conducted for consecutive lengths of history to test the hypothesis that the mean difference between fractions of correct choices is equal to zero. From the specific length of history for which the hypothesis cannot be neglected any more, the provider's decision cannot be improved significantly any more.

Figure 2 implies that an increasing length of history increases the fraction of correct choices. This implication is tested by means of an analysis of variance. The hypothesis that the mean fraction of correct choices is equal for different lengths of history can be neglected according to the results of the analysis of variance as is shown in Table 4. Furthermore, Figure 2 reveals that the marginal improvement in the fraction of correct choices decreases with an increasing history. This allows to identify the threshold value for the required amount of monitoring data as the amount of data from which on the fraction of correct choices cannot be improved significantly any more. This is facilitated

by exerting two-sided unpaired t-tests on the mean fractions of correct choices for sequential lengths of history, i.e. average fraction of correct choices for history of length 100 and average fraction of correct choices for history of length 50. The hypothesis that the means of fraction of correct choices are equal is tested to confidence level of 5%. As a result of this test, a length of 20 observations per SLA portfolio is identified as the threshold value as the Null hypothesis cannot be rejected any more (compare to the Significance Value in Table 5).

	Sum of squares	Mean Square	F	Sig.
Between Groups	14.269	1.019	14.763	.000
Within Groups	154.02	.069		
Total	168.571			

Table 4 Analysis of Variance: Fraction of Correct Choices

	t	Sig.
Diff. 10 to 20	-2.413	.016
Diff 20 to 30	-1.142	.254

Table 5 Results of pairwise two-sided unpaired t-tests

7 Conclusion

In this paper, the model of Agreement Networks was proposed and a decision support model for service providers that enables the selection of SLA combinations was introduced that allows for the provider's risk-awareness. Risk of SLA violation in this context is expressed analogously to security portfolio risk stemming from finance. This approach enables a service provider to select the combination of SLAs based upon past observations that is expected to incur the lowest risk of SLA violation on the provider while meeting a pre-defined expected profit constraint. Note that our approach might reject combinations of SLAs, which combine above average expected profit with only slightly higher variance. To avoid this inefficiency of choice, the trade-off between risk and expected profit in the provider's preferences could be explicitly taken into account, e.g. by applying a utility function as an objective function.

The approach was evaluated by investigating the required amount of available monitoring data and identified and suggests 20 observations for each SLA combination. This amount of data is required in case that all monitoring data is treated equal and no preferences for the age of the monitoring data are included. In order to investigate the impact of changing violation probability distributions that could be caused e.g. by an extension of the infrastructure, the measure of risk has to be adapted in order to weight more recent data higher than older data. This could be achieved by an approach like exponential smoothing (Trigg and Leach 1967).

Further future work includes the evaluation of an extension of the risk-aware decision support in a scenario with a service intermediary. The intermediary on the one hand has to make a decision analogously to the one presented in the work at hand about the SLAs that he wants to offer to consumers. On the other hand, the intermediary does not provide all of the services by themselves but requests services from providers and hence, has to select providers and SLAs for those services.

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